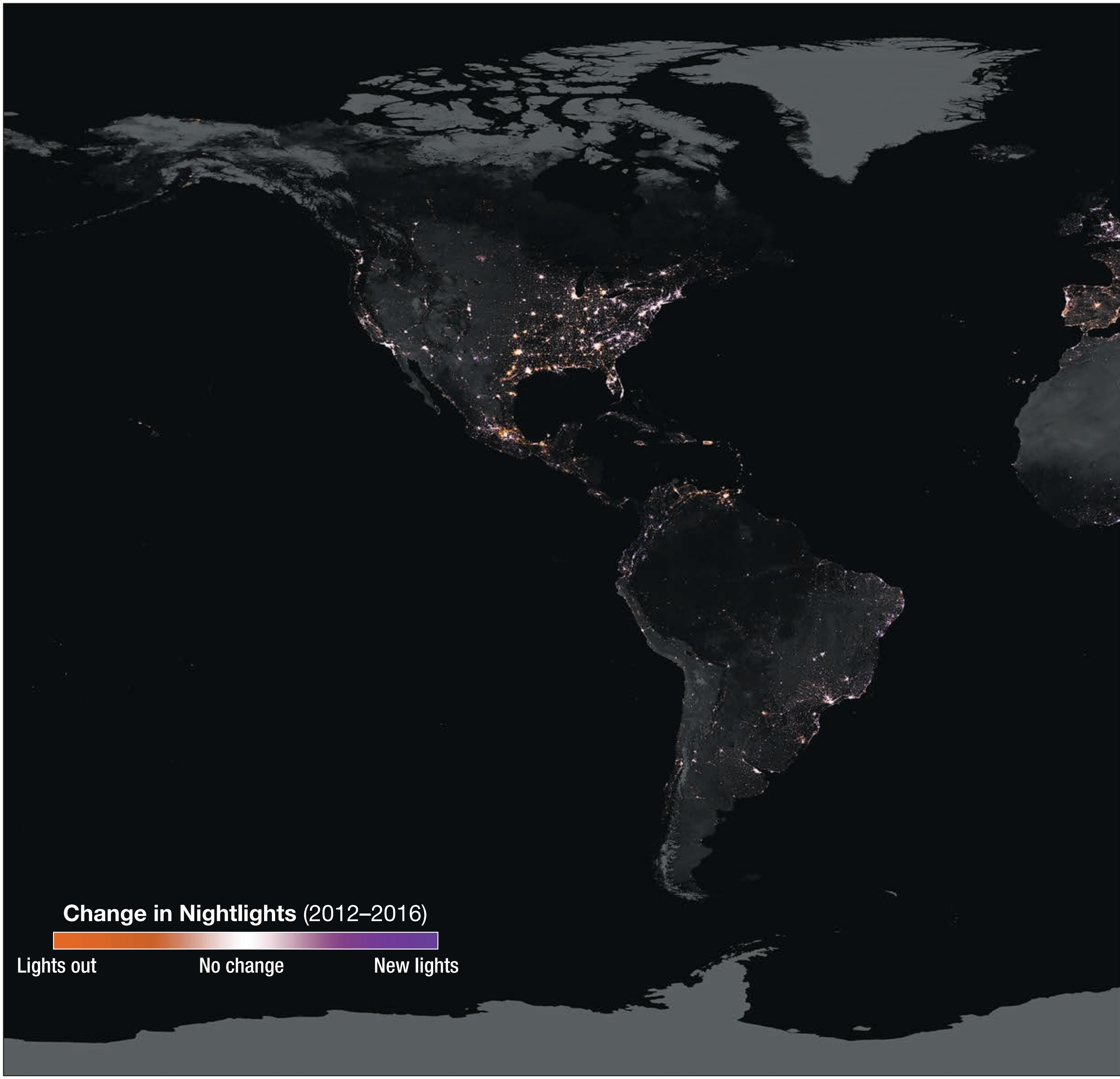
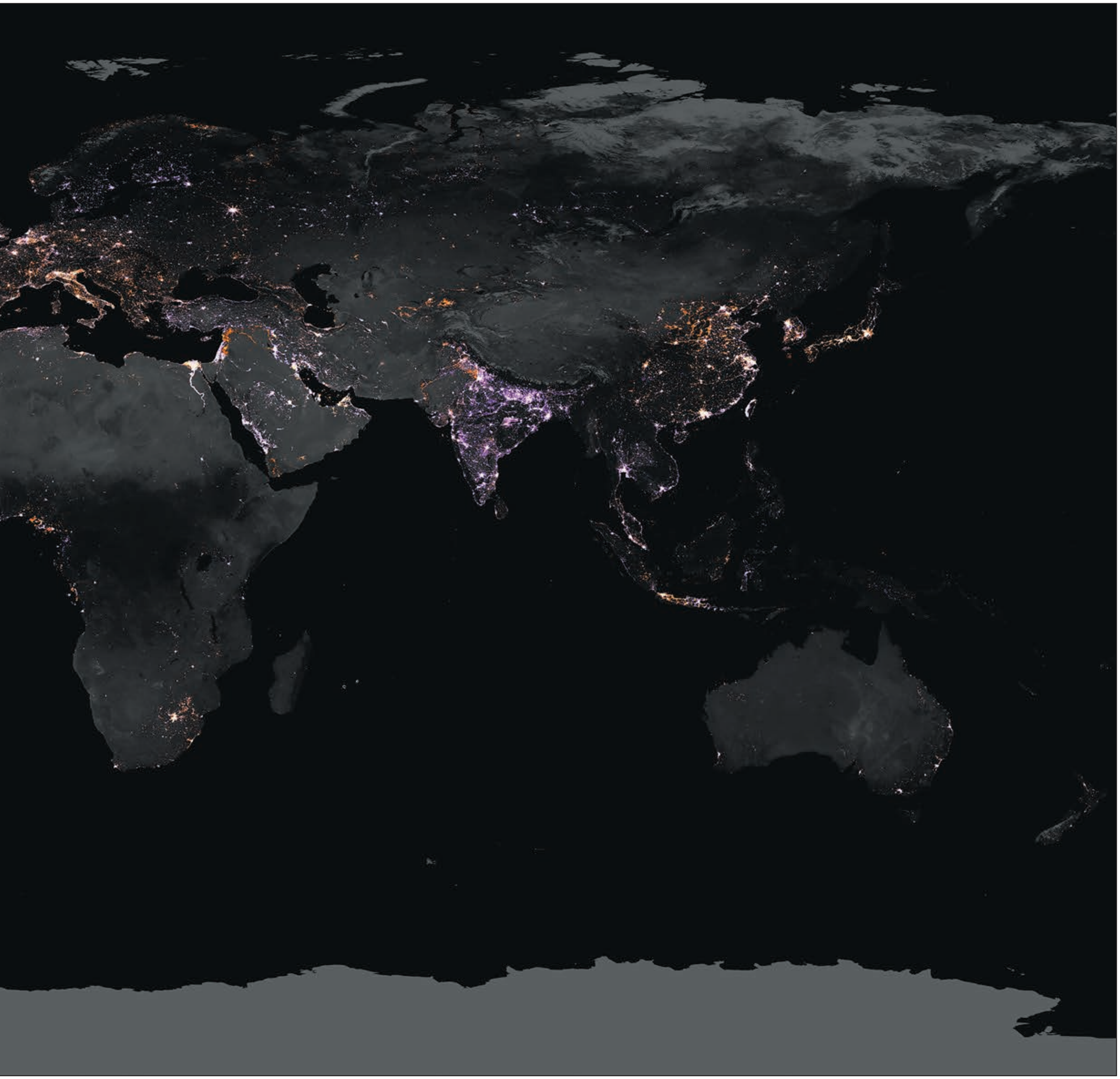
New Orleans, at the mouth of the Mississippi River, to Brownsville, Texas, in the westernmost Gulf. In recent years a new pattern of lights has appeared on the landscape and revealed the oil- and gas-production zone of south-central Texas. This long, less-dense swath of pinpoints spreads across 210 miles (330 kilometers) of what is now known as shale-fracking country.

A World of Change

This map shows the change in lighting intensity from 2012 to 2016. The map was created using two separate nightlight datasets (from 2012 and 2016) derived using data from the VIIRS instrument on Suomi NPP. Dark purple represents areas with new light since 2012, while dark orange represents



areas where light existed in 2012 but no longer exists in 2016. Areas where lighting intensity stayed the same between 2012 and 2016 appear white. Varying shades of purple and orange indicate areas that have become brighter or dimmer since 2012, respectively. To view and learn about a few regional examples up close, see stories on pages 114–115 and 140–141.



Nature's Light Shows

At night, with the light of the Sun removed, nature's brilliant glow from Earth's surface becomes visible to the naked eye from space. Some of Earth's most spectacular light shows are natural, like the aurora borealis, or Northern Lights, in the Northern Hemisphere (aurora australis, or Southern Lights, in the Southern Hemisphere). The auroras are natural electrical phenomena caused by charged particles that race from the Sun toward Earth, inducing chemical reactions in the upper atmosphere and creating the appearance of streamers of reddish or greenish light in the sky, usually near the northern or southern magnetic pole. Other natural lights can indicate danger, like a raging forest fire encroaching on a city, town, or community, or lava spewing from an erupting volcano.

Whatever the source, the ability of humans to monitor nature's light shows at night has practical applications for society. For example, tracking fires during nighttime hours allows for continuous monitoring and enhances our ability to protect humans and other animals, plants, and infrastructure. Combined with other data sources, our ability to observe the light of fires at night allows emergency managers to more efficiently and accurately issue warnings and evacuation orders and allows firefighting efforts to continue through the night. With enough moonlight (e.g., full-Moon phase), it's even possible to track the movement of smoke plumes at night, which can impact air quality, regardless of time of day.

Another natural source of light at night is emitted from glowing lava flows at the site of active volcanoes. Again, with enough

[Above] Astronauts onboard the International Space Station are treated to a spectacular display of the aurora australis while in orbit over Southern New Zealand on September 17, 2011.

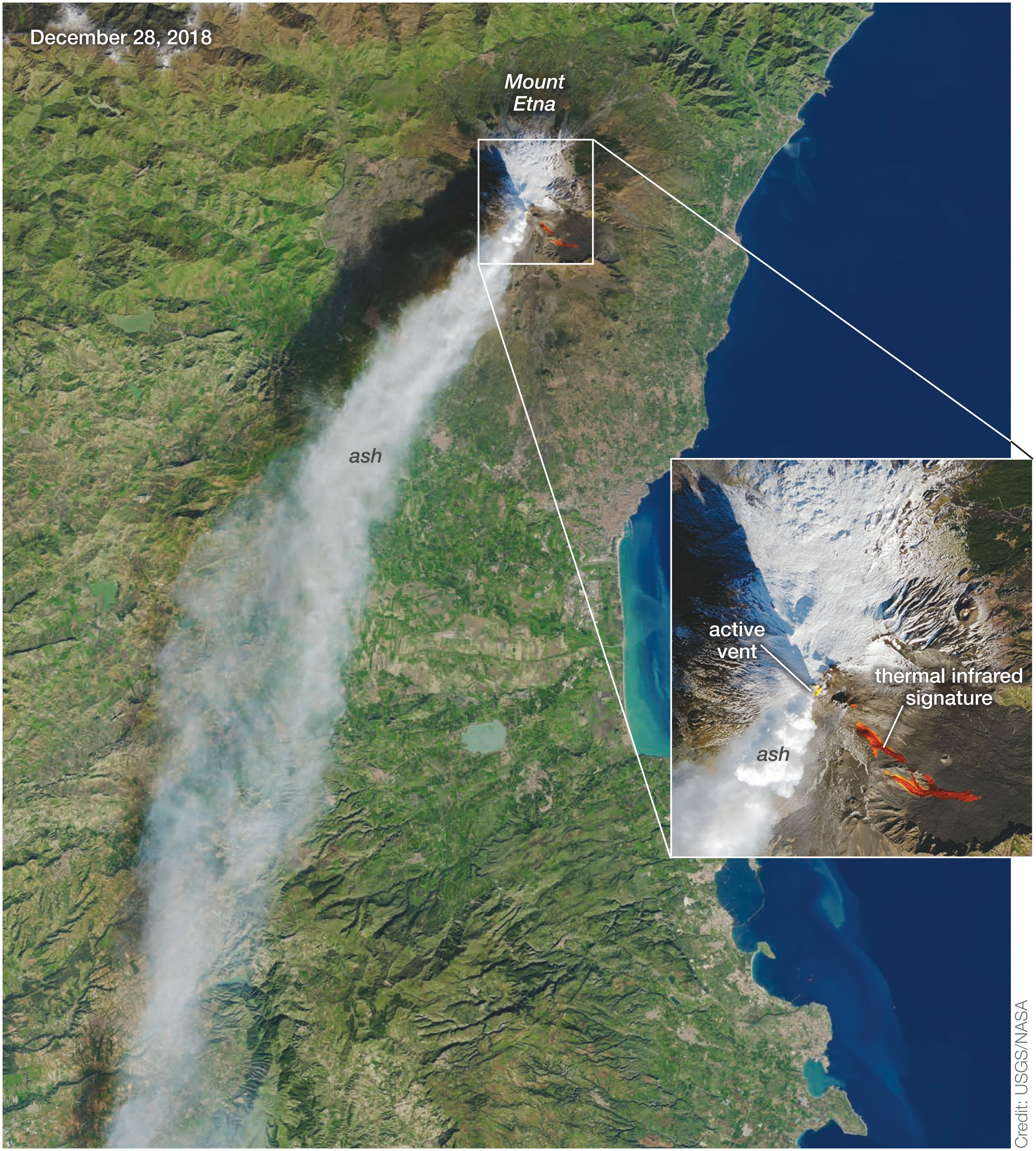
[Opposite page] For the first time in perhaps a decade, Mount Etna experienced a "flank eruption"—erupting from its side instead of its summit—on December 24, 2018. The activity was accompanied by 130 earthquakes occurring over three hours that morning. Mount Etna, Europe's most active volcano, has seen periodic activity on this part of the mountain since 2013. The Operational Land Imager (OLI) on the Landsat 8 satellite acquired the main image of Mount Etna on December 28, 2018. The inset image highlights the active vent and thermal infrared signature from lava flows, which can be seen near the newly formed fissure on the southeastern side of the volcano. The inset was created with data from OLI and the Thermal Infrared Sensor (TIRS) on Landsat 8. Ash spewing from the fissure cloaked adjacent villages and delayed aircraft from landing at the nearby Catania airport. Earthquakes occurred in the subsequent days after the initial eruption and displaced hundreds of people from their homes. For nighttime images of Mt. Etna's March 2017 eruption, see pages 48-51.

moonlight, we can even track volcanic ash plumes, which can be hazardous to airplanes in flight. The ash, made up of tiny pieces of glass and rock, is abrasive to engine turbine blades and can melt on the blades and other engine parts, causing damage and even engine stalls, with consequent danger to the plane's integrity and passenger safety. Volcanic ash also reduces visibility for pilots and can cause etching of windshields, further reducing pilots' ability to see. Nightlight images can be combined with thermal images to give a more complete view of volcanic activity on Earth's surface, as will be seen later.

The VIIRS DNB takes advantage of moonlight, airglow (the atmosphere's self-illumination through chemical reactions), zodiacal light (sunlight scattered by interplanetary dust), and starlight from the Milky Way. By using these dim light sources, the DNB can detect changes in clouds, snow cover, and sea ice. Geostationary Operational Environmental Satellites (GOES), managed by the National Oceanic and Atmospheric Administration (NOAA), orbit over Earth's equator, offering uninterrupted observations of North America, whereas high-latitude areas such as Alaska benefit from polar-orbiting satellites like Suomi NPP. Polar-orbiting satellites provide significant overlapping coverage at the poles, enabling more data to be acquired in these regions. For example, during polar darkness (i.e., winter months), VIIRS DNB data allow scientists to observe sea ice formation and snow cover extent at the highest latitudes, in addition to spotting where there is clear water for ships to pass through.

The use of nightlight data by weather forecasters is growing as the VIIRS instrument can be used to see clouds at night when they are lit by moonlight and lightning. Scientists use nightlight data to study nighttime behavior of weather systems, including severe storms, which can develop and strike populous areas at night as well as during the day. Combined with thermal data, visible nightlight data can also be used to see clouds at various heights in the atmosphere at night, such as dense marine fog, which allows weather forecasters to issue marine advisories with higher confidence than ever before and, therefore, greater utility—see "Marine Layer Clouds—California" on page 56.

In this section of the book you will see how nightlight data are used to observe nature's spectacular light shows across a wide range of sources.



Credit: USGS/NASA

Fires

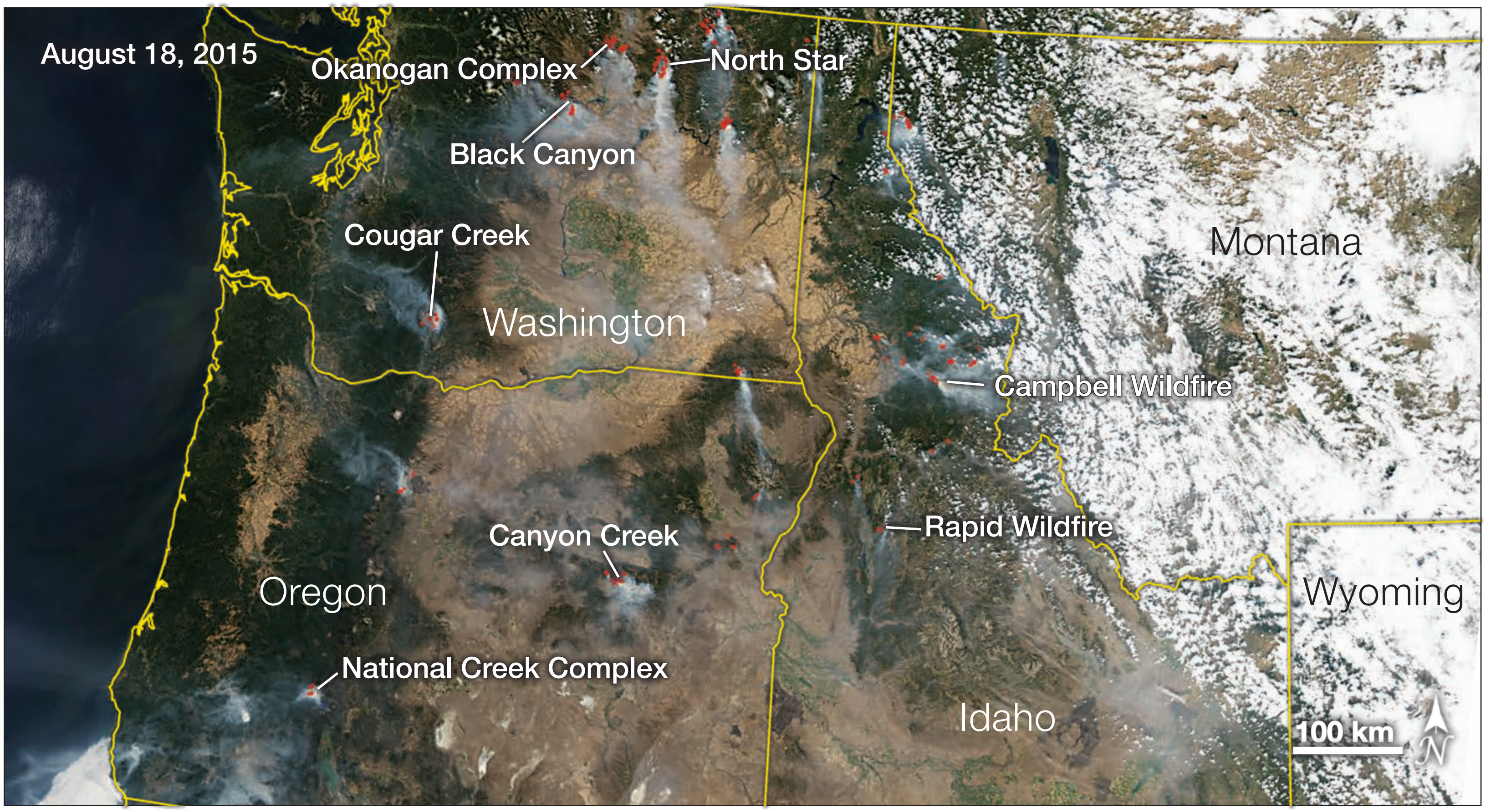
Tracking Fires Day and Night—Northwest United States

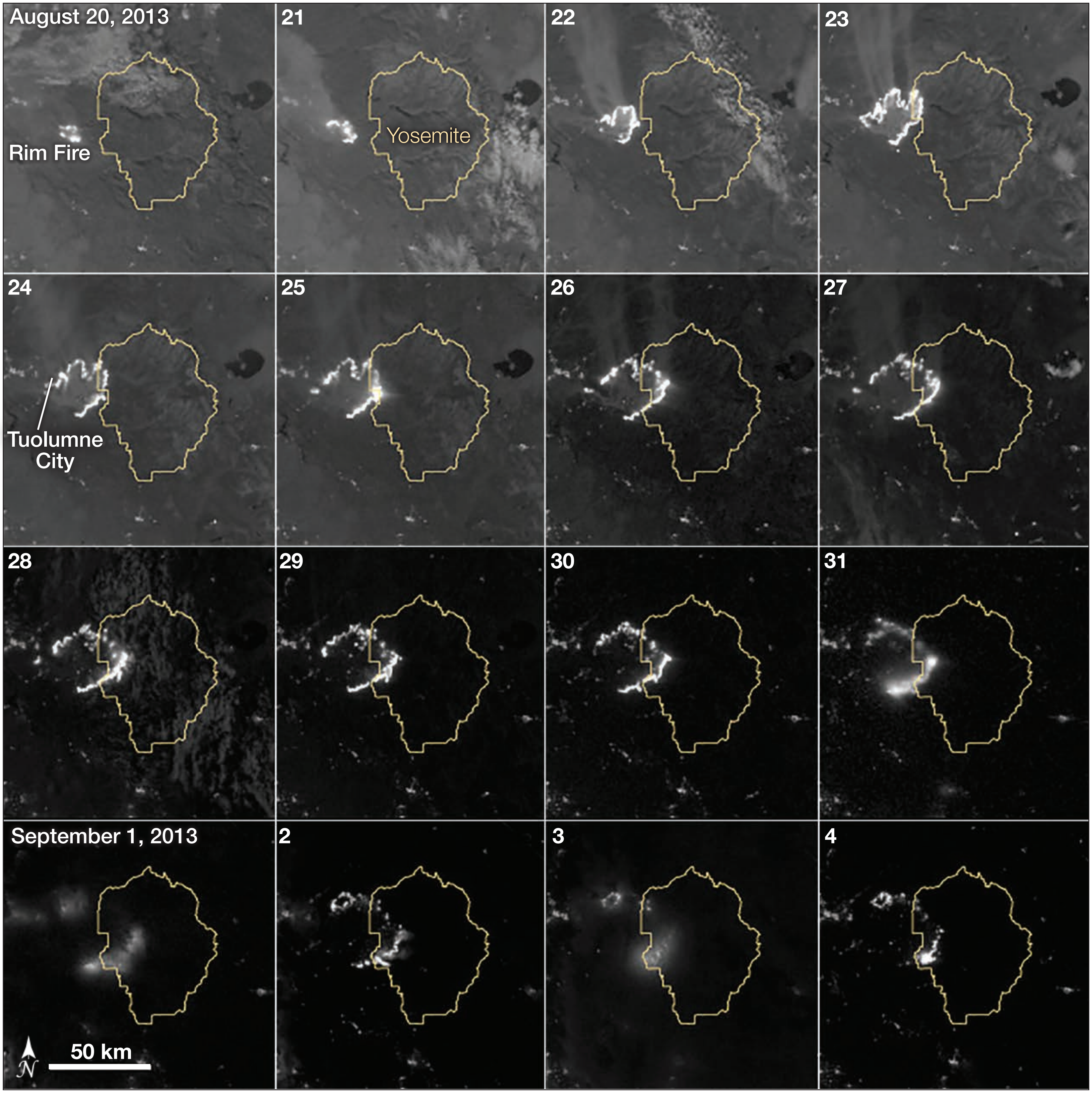
In summer 2015, wildfires raged across the western contiguous United States and Alaska. Many of those fires burned in the Northwest United States, visible in these images from late August. The nighttime image (top, opposite) was acquired in the early morning (local time) on August 19, 2015, by the VIIRS DNB on the Suomi NPP satellite. Labels point to the large, actively burning fires in the region.

The daytime image (bottom, opposite) shows the same area in natural color, acquired during the afternoon on August 18, 2015, with the Moderate Resolution Imaging Spectroradiometer (MODIS) on NASA's Aqua satellite. Red outlines indicate hot spots where MODIS detected unusually warm surface temperatures, generally associated with fires. Thick plumes of smoke are visible emanating from the hot spots.

According to the Northwest Interagency Coordination Center, the Okanogan Complex Fire in Washington was among the larger active fires; as of August 20, the fire had burned 91,314 acres (143 square miles, or 370 square kilometers). In Oregon, the Canyon Creek Complex Fire had burned 48,201 acres (75 square miles, or 195 square kilometers), destroyed 26 residences, and threatened another 500. Both fires were less than 40 percent contained at the time the images were acquired.


August 19, 2015
Okanogan Complex
North Star
Black Canyon
Seattle
Spokane
Cougar Creek
Washington
Montana
Campbell Wildfire
Portland
Canyon Creek
Rapid Wildfire
Oregon
Wyoming
National Creek Complex
Boise
Idaho
100 km
N


August 18, 2015
Okanogan Complex
North Star
Black Canyon
Cougar Creek
Montana
Washington
Campbell Wildfire
Rapid Wildfire
Canyon Creek
Oregon
Wyoming
National Creek Complex
Idaho
100 km
N


August 20, 2013
21
22
23
Rim Fire
Yosemite
24
25
26
27
Tuolumne City
28
29
30
31
September 1, 2013
2
3
4
N
50 km